INCH-POUND
MIL-DTL-24643/13D
22 August 2002
SUPERSEDING
MIL-C-24643/13C
14 March 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LSTTRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor - AWG size No. 20, class B stranding, uncoated.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Nominal wall thickness 0.020 inch; wall thickness may vary from nominal as necessary, in order to meet the specified electrical requirements. One black and one white or natural conductor for each pair.
Fourth -	Two conductors, one black and one white or natural conductor, cabled together to form a pair. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests.
Fifth -	Braided shield of copper, uncoated or tin coated. Minimum coverage of 85 percent.
Sixth -	Shield insulation of two polyester tapes per NEMA FI-4, outer tape sealed. An alternate shield insulation may consist of one polyester tape per NEMA FI-4 plus an extruded jacket of a suitable transparent material, nominal thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Construction (non-watertight) (continued)

Seventh - The specified number of pairs (see table I) shall be cabled together with left hand lay. The length of lay shall be determined to permit a completed cable to meet the twisting and bending endurance tests. Cabling sequence shall be consecutive, starting with pair number 1, from center outward. A single pair shall not be used as center for the cable core. Fillers shall be used in the center and valleys, as necessary, to form a firm, rounded assembly.
Eighth - Binder tape applied helically with overlap.
Ninth - Cross-linked polyolefin jacket.

TABLE I. Details.

Military part no. M24643/13	Size	Number of shielded pairs	Jacket wall thickness (min avg) (inches)	Overall diameter (inches) Type LSTTRS	
				min	max
-01UN	2	2	0.050	0.629	0.680
-02UN	4	4	0.050	0.685	0.740
-03UN	6	6	0.050	0.814	0.880
-04UN	8	8	0.050	0.916	0.990
-05UN	10	10	0.050	0.999	1.080
-06UN	12	12	0.070	1.017	1.100
-07UN	16	16	0.070	1.100	1.190

EXAMINATION AND TESTS:

Basic electrical: Requirements:
Conductor resistance - ohms/1000 feet at 25°C, maximum 11.95
Voltage withstand - volts, root mean square, minimum
conductor to conductor 2000
shield to shield 200
Insulation resistance - megohms -1000 feet, minimum
Conductor to shield 500
Shield to shield 100
Conductor and shield continuity ..... No failure

Group A:
Visual and dimensional ..... No failure
Mutual capacitance - pF/ft at 1 megahertz (MHz), maximum 28
Capacitance unbalance - percent maximum 10
Characteristic impedance - ohms, at 1 MHz minimum 75
maximum 90

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Group B:
- Cross-linked proof test (percent, maximum)
  - Insulation 50
- Jacket (When tested at 200°C) 50
- Tear - pounds per inch thickness, minimum
  - (ASTM D 470) 35

Physicals (unaged)
- Insulation
  - Tensile strength - lb/in^2, minimum 1800
  - Elongation - percent, minimum 250
- Jacket (cable)
  - Tensile strength - lb/in^2, minimum 1300
  - Elongation - percent, minimum 160
- Twisting and bending endurance -
  - At 75°C, cycles, minimum 6000
  - At minus 30 ± 2°C, cycles, minimum 2000

NOTE: 1. Roller diameter for bending endurance test shall be as follows:

Cable type and size	Roller diameter (inches, maximum)
LSTTRS-2	2.0
LSTTRS-4	2.2
LSTTRS-6	3.0
LSTTRS-8	3.5
LSTTRS-10	4.0
LSTTRS-12	4.0
LSTTRS-16	4.5

Group C: Requirements:

Physicals (aged)
- Insulation
  - Air oven
    - Tensile strength - percent of unaged, minimum 80
    - Elongation - percent of unaged minimum 80
  - Jacket (cable)
    - Air oven
      - Tensile strength - percent of unaged, minimum 60
      - Elongation - percent of unaged minimum 60
  - Hot oil immersion
    - Tensile strength - percent of unaged, minimum 50
    - Elongation - percent of unaged minimum 50
- Shrinkage ..... No failure
- Permanence of printing (jacket) - cycles, minimum 125
- Heat distortion - percent of unaged, maximum 30

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Group D:

Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ± 5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Shield covering 2
Halogen content - percent, maximum Jacket 0.2
Fillers 0.2
Immersion (jacket)
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Smoke index, maximum
Jacket 25
Fillers 45
Insulation 45
Toxicity index, maximum
Jacket 5
Fillers 5
Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket)..... No failure

UNIT ORDERING LENGTHS:
All sizes 1500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-011)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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